RIDE ANALYTICS REPORT

Bykea Ride Analysis — Bookings, Revenue & Operational Performance

Power BI Dashboard & Data Analytics Project

PREPARED BY

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REPORTING PERIOD

January – December 2024

DATASET

Bykea Ride Bookings Dataset (153,000 raw booking records, Karachi)

REPORT TYPE

Booking, Revenue & Service-Quality Performance Analysis

Executive Summary

This report analyzes 153,000 ride-booking records from Bykea's Karachi operations across 2024 to understand booking volume, completion performance, revenue drivers, cancellations, and service quality. The dataset covers seven vehicle types (Bike, Rickshaw, Chingchi, Suzuki Mehran, Toyota Corolla, Honda Civic, and Toyota Hiace) across hundreds of Karachi pickup and drop points. The analysis is designed to help reduce cancellations, grow revenue in underperforming time windows, and strengthen service quality and customer retention.

Business Overview

Of the 153,000 bookings raised in 2024, 94,860 (62.00%) were completed, generating total revenue of Rs 47.26M at an average booking value of Rs 508. The remaining bookings were split across driver cancellations (18.01%), customer cancellations (7.00%), no driver found (6.99%), and incomplete rides (6.01%). Riders travelled an average of 26.0 km per completed trip, with an average driver arrival time (VTAT) of 8.46 minutes and average ride duration (CTAT) of 29.15 minutes. Overall performance is healthy, with the clearest opportunity lying in reducing driver-side cancellations, which are more than double any other loss category.

KPI Summary

KPI	Value
Total Bookings Raised	**153,000**
Completed Bookings	**94,860 (62.00%)**
Total Revenue (Completed Rides)	**Rs 47.26M**
Average Booking Value	**Rs 508**
Completion Rate	**62.00%**
Driver Cancellation Rate	**18.01% (27,551 rides)**
Customer Cancellation Rate	**7.00% (10,705 rides)**
No Driver Found Rate	**6.99% (10,695 rides)**
Incomplete Ride Rate	**6.01% (9,189 rides)**
Avg. Driver Arrival Time (VTAT)	**8.46 min**
Avg. Ride Duration (CTAT)	**29.15 min**
Revenue per KM	**Rs 19.54**
Service Quality Score	**86.4 / 100**

Key Findings

- **Booking Outcome:** 62.00% of bookings completed successfully; driver-side cancellations (18.01%) are by far the largest single loss category — more than driver-no-show, customer cancellation, and incomplete rides combined.
- **Vehicle Mix:** Rickshaw is the leading vehicle by both volume (38,183 bookings) and revenue (Rs 12.88M, 27% of total), followed by Suzuki Mehran and Toyota Corolla.

• **Payment Behaviour:** Digital payments (EasyPaisa, JazzCash, Debit/Credit Card) account for 75.1% of completed-ride revenue versus 24.9% for cash, with EasyPaisa alone contributing Rs 21.27M.

• **Time-of-Day:** Morning bookings generate the highest revenue (Rs 14.39M), closely followed by Evening (Rs 13.86M) — the two peak windows together drive 60% of daily revenue.

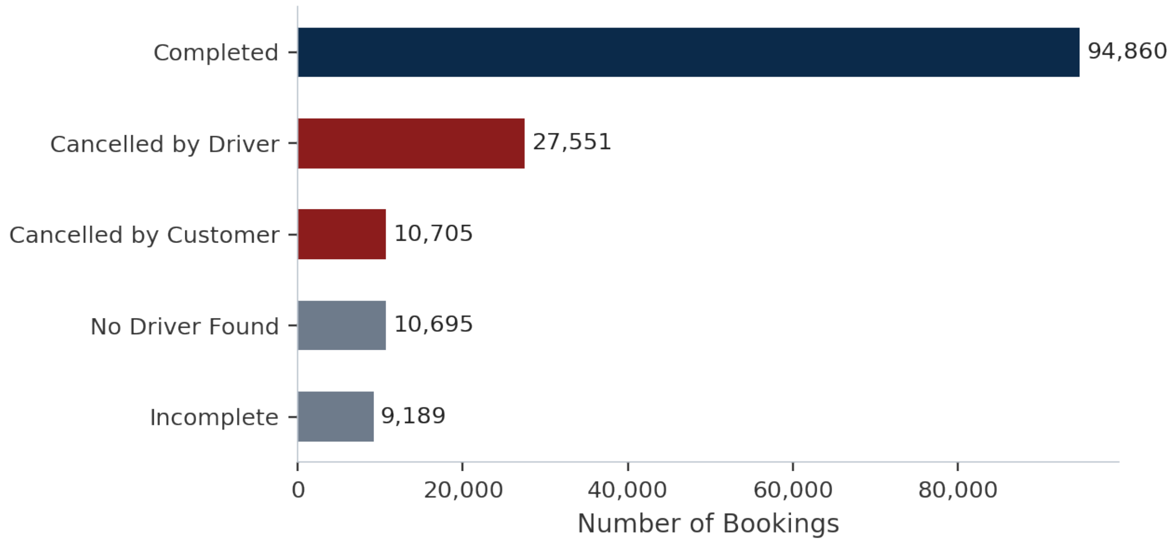
• **Retention Gap:** Only 2.45% of the 92,547 unique completed-ride customers booked more than once in the year, pointing to a significant repeat-usage opportunity.

• **Service Quality:** Average driver rating (4.23/5) and customer rating (4.40/5) combine into a Service Quality Score of 86.4/100, consistent across all seven vehicle types.

Detailed Business Analysis

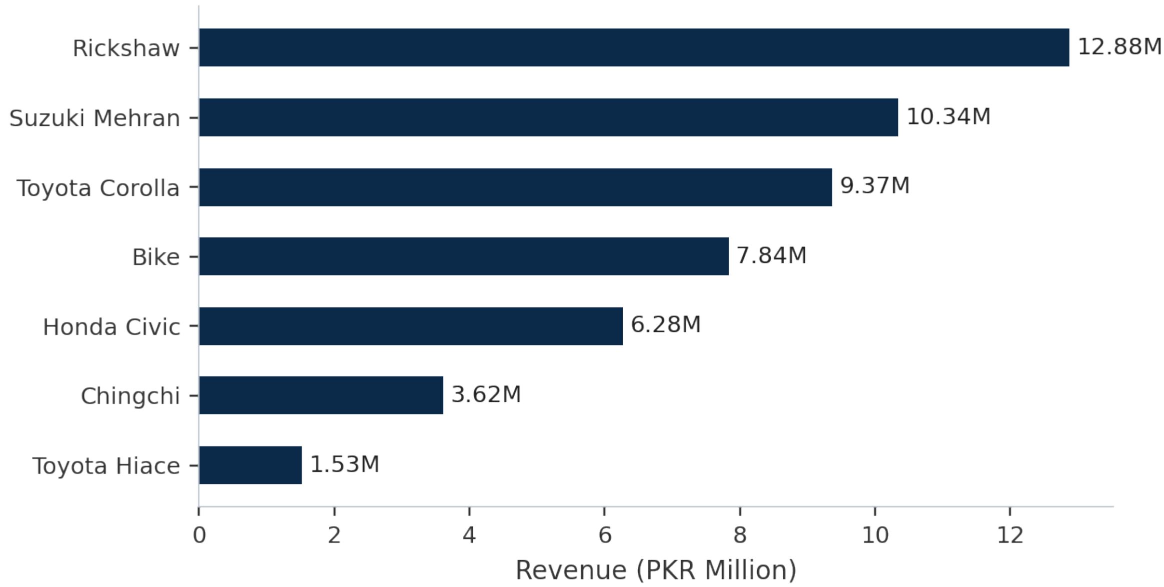
Booking Status & Completion Analysis

Of 153,000 total bookings, 94,860 (62.00%) completed successfully. Driver-related cancellations account for 27,551 rides (18.01%) — the single largest source of lost bookings, nearly double the next-largest category. Customer cancellations (10,705, 7.00%) and no-driver-found outcomes (10,695, 6.99%) are close to each other in size, while 9,189 rides (6.01%) were marked incomplete after starting.



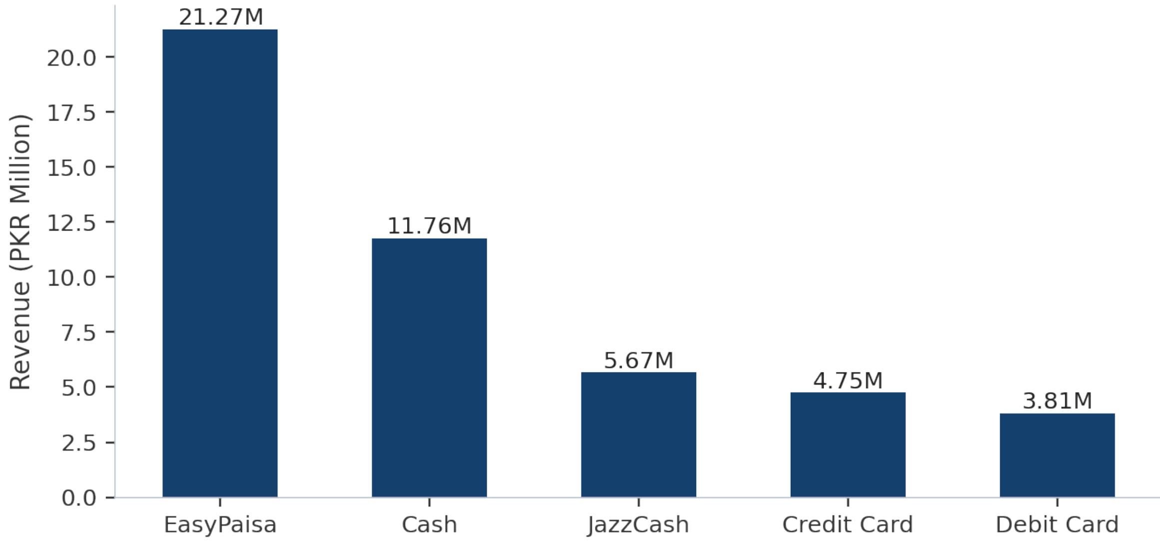
Vehicle Type Analysis

Rickshaw is the dominant vehicle type, contributing Rs 12.88M (27.2%) of total revenue across 38,183 bookings, followed by Suzuki Mehran (Rs 10.34M) and Toyota Corolla (Rs 9.37M). Toyota Hiace, used for larger/shared rides, contributes the least at Rs 1.53M but also carries the fewest bookings (4,533), suggesting it is a niche rather than core offering. Average trip distance is virtually identical (~24.4–25.0 km) across all vehicle types, indicating vehicle choice is driven by price point and availability rather than trip length.


Rickshaw
Suzuki Mehran
Toyota Corolla
Bike
Honda Civic
Chingchi
Toyota Hiace
12.88M
10.34M
9.37M
7.84M
6.28M
3.62M
1.53M
0
2
4
6
8
10
12
Revenue (PKR Million)

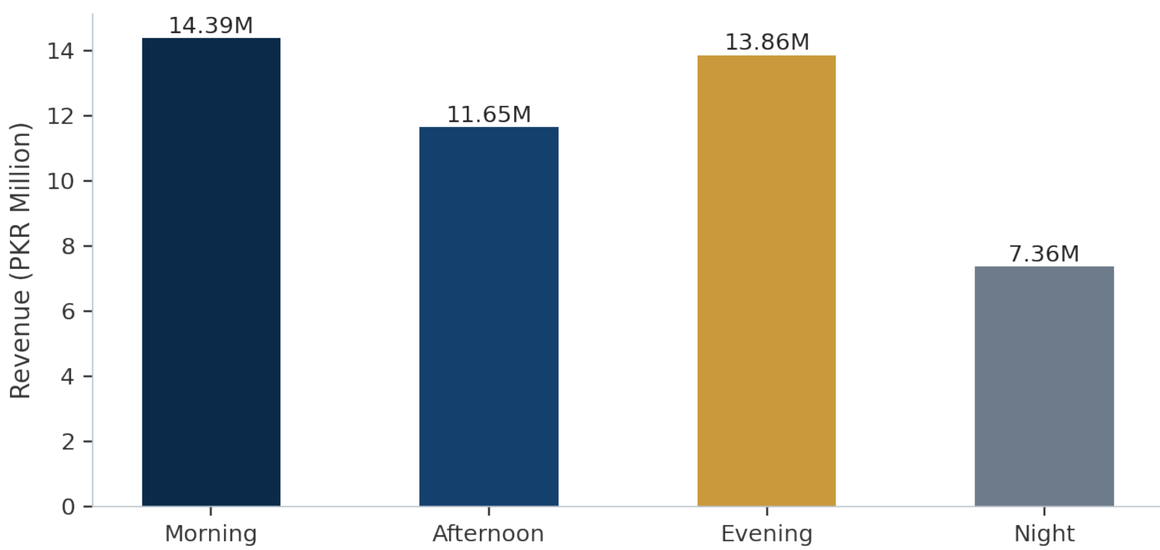
Payment Method Analysis

EasyPaisa is the leading payment method by revenue (Rs 21.27M), followed by Cash (Rs 11.76M), JazzCash (Rs 5.67M), Credit Card (Rs 4.75M) and Debit Card (Rs 3.81M). Digital wallets and cards together account for 75.1% of completed-ride revenue, confirming that Bykea's customer base is now predominantly cashless — an important signal for prioritizing digital payment reliability and fraud controls over cash-handling processes.



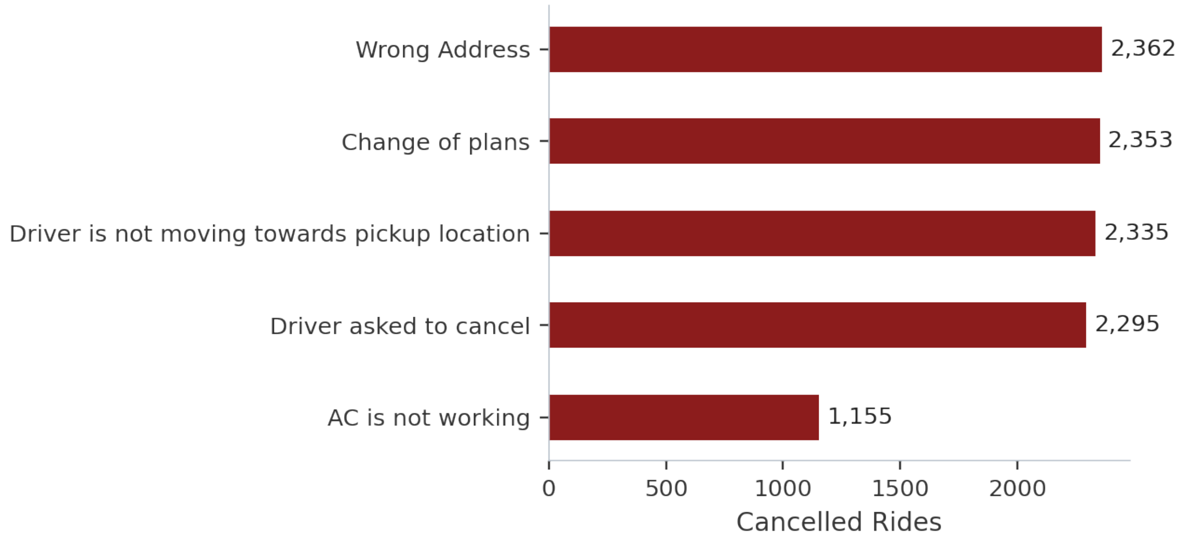
Time-of-Day (Day Part) Analysis

Bookings are heaviest in the Morning window (46,323 bookings, Rs 14.39M revenue) and Evening window (45,015 bookings, Rs 13.86M revenue) — together these two peak periods generate 60% of total revenue. Afternoon trails at Rs 11.65M, while Night is the smallest window at Rs 7.36M, reflecting Karachi's typical commute-driven demand pattern around morning and evening rush hours.

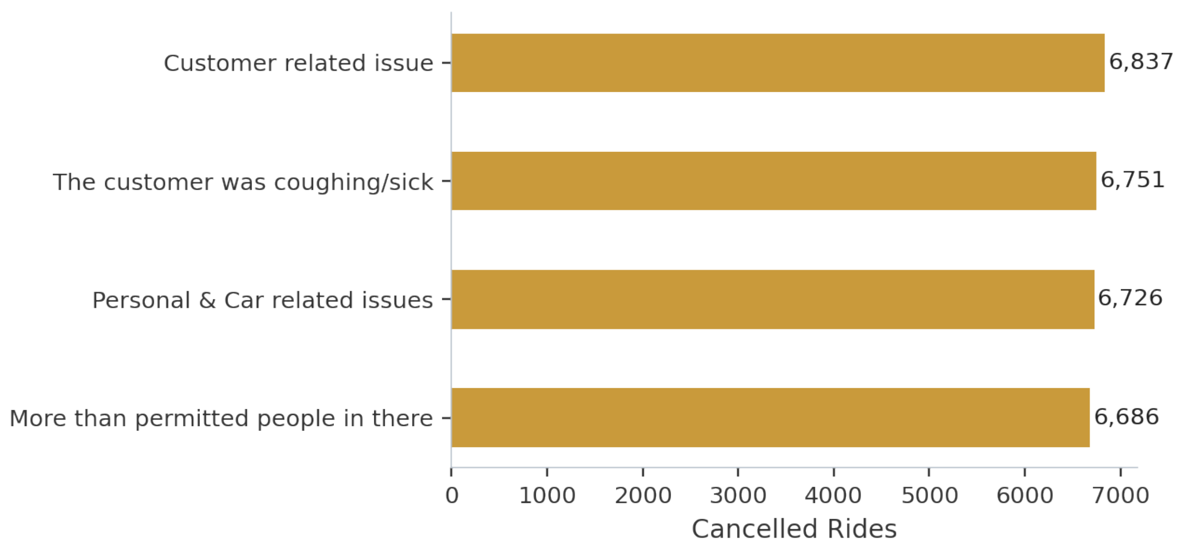


Cancellation Analysis

Customer-side cancellations are fairly evenly spread across four reasons: Wrong Address (2,362), Change of Plans (2,353), Driver Not Moving Towards Pickup (2,335), and Driver Asked to Cancel (2,295) — the latter two together (4,630 rides) point to driver-behaviour issues that surface even in customer-attributed cancellations. AC Not Working (1,155) is a smaller but easily fixable vehicle-maintenance driven reason.



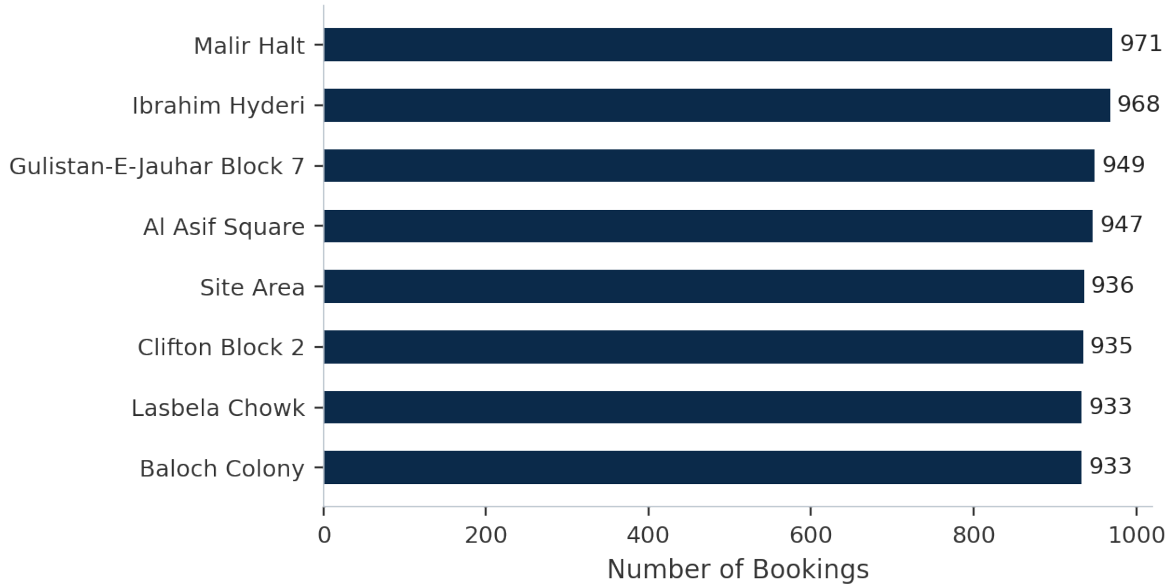
Driver-side cancellations — the largest single loss category overall — are led by Customer Related Issue (6,837), Customer Coughing/Sick (6,751), Personal & Car Related Issues (6,726), and More Than Permitted People (6,686). These four reasons are almost evenly distributed, suggesting no single root cause dominates; a combination of stricter passenger-count enforcement, driver health-screening policy clarity, and vehicle-maintenance support would address most of this category.



Incomplete rides (9,189 total) are driven by Customer Demand (3,040), Vehicle Breakdown (3,012), and Other Issue (2,948) in roughly equal measure — Vehicle Breakdown in particular is a direct signal for tightening vehicle condition checks, especially for older Rickshaw and Chingchi units.

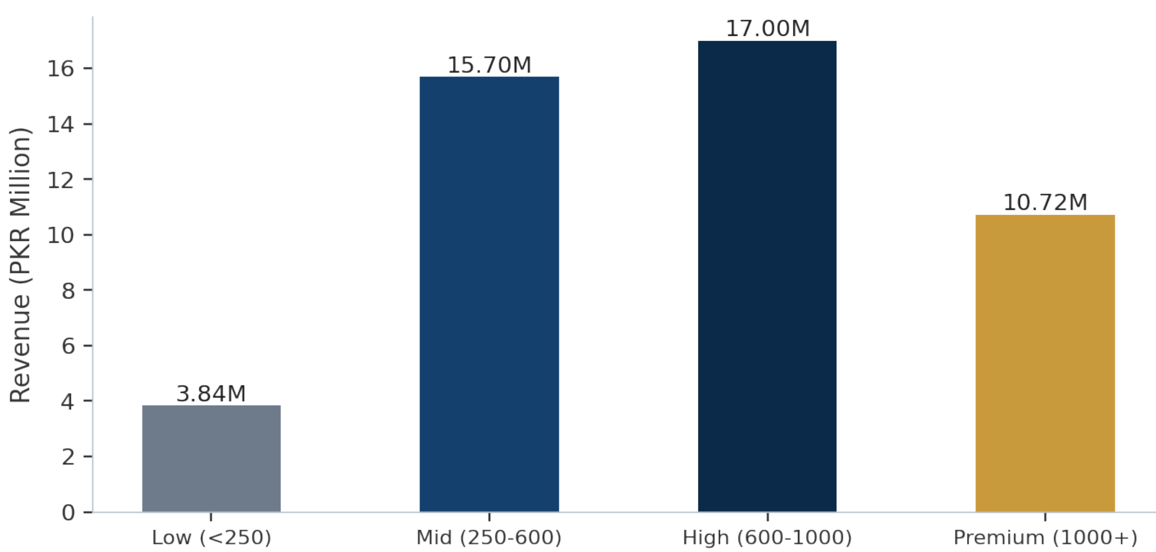
Location Hotspot Analysis

Pickup demand is spread broadly across Karachi with no single overwhelming hotspot — the busiest pickup points are Malir Halt (971 bookings), Ibrahim Hyderi (968), Gulistan-e-Jauhar Block 7 (949), Al Asif Square (947), and Site Area (936). This even spread across more than 150 distinct pickup areas suggests demand is genuinely city-wide rather than concentrated in a few commercial districts, which has implications for driver-supply positioning across zones rather than centralizing fleet in one area.



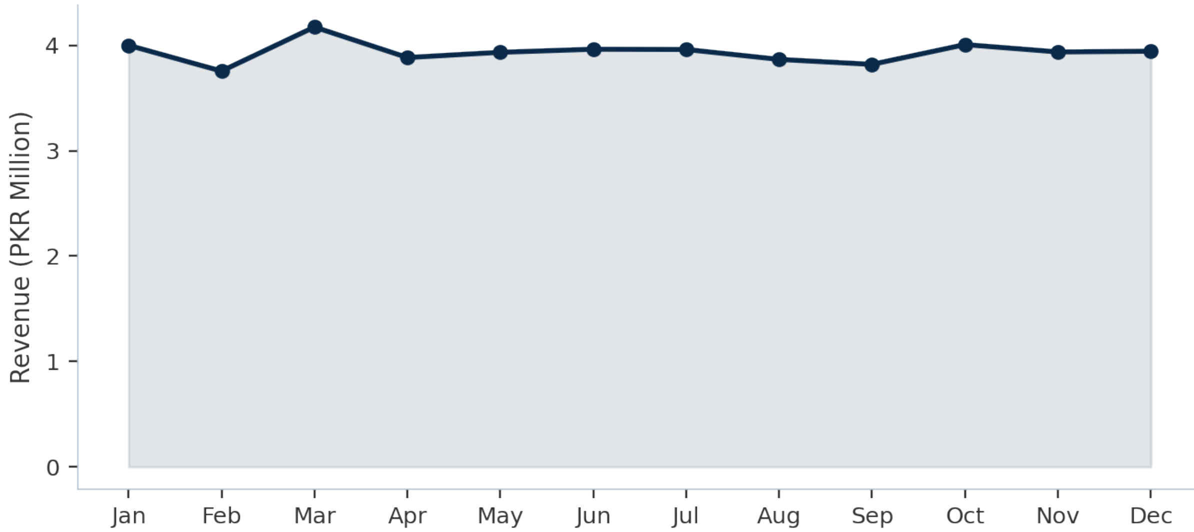
Revenue Tier Analysis

Mid-value rides (Rs 250–600) are the largest contributor to completed-ride volume (38,788 rides, Rs 15.70M) and High-value rides (Rs 600–1,000) contribute the most revenue per booking, generating Rs 17.00M from 21,634 rides. Premium rides (Rs 1,000+) are a smaller segment (7,305 rides) but still contribute Rs 10.72M (22.7% of revenue) — and 59.6% of these premium bookings occur during Morning or Evening peak windows, indicating premium demand is closely tied to commute urgency rather than leisure travel.



Monthly Trend Analysis

Completed-ride revenue is remarkably stable across 2024, ranging narrowly between Rs 3.76M (February) and Rs 4.17M (March). There is no strong seasonal spike or trough — the pattern suggests demand is driven by consistent day-to-day commuting needs rather than seasonal events, which supports predictable driver-supply and revenue planning throughout the year.



Customer Retention & Service Quality

Of 92,547 unique customers with at least one completed ride in 2024, only 2,268 (2.45%) booked more than once — a low repeat-usage rate for a city-wide ride service and the single biggest growth lever available without acquiring new customers. Service quality itself is not the barrier: the combined driver and customer rating average translates to a Service Quality Score of 86.4/100, and this holds consistently across every vehicle type (4.23–4.24 average driver rating, 4.40–4.41 average customer rating). This points toward retention/loyalty mechanics — not service quality — as the priority fix.

Recommendations

1. Cancellation Reduction

• Investigate driver-side cancellations first — at 18.01% they are the largest loss category; start with the four near-equal reasons (customer issue, health-related, personal/car issues, passenger-count) rather than assuming one dominant cause.

• Enforce and communicate passenger-count limits at booking time to pre-empt 'more than permitted people' cancellations.

• Introduce a pre-ride vehicle condition checklist for Rickshaw and Chingchi drivers to reduce Vehicle-Breakdown-driven incomplete rides.

2. Revenue Growth

• Prioritize Morning and Evening peak-window driver availability, since these two windows already drive 60% of revenue and show the highest Premium-tier concentration (59.6%).

• Test targeted pricing or bundled offers for the Mid-value (Rs 250–600) segment, the largest volume tier, to nudge more rides into the High-value tier.

• Expand Toyota Hiace availability selectively in high-demand zones — it is currently the smallest fleet segment but may be underutilized rather than genuinely low-demand.

3. Customer Retention

• Launch a repeat-rider incentive (e.g. loyalty credit after the 2nd/3rd ride) — only 2.45% of customers currently rebook, the largest single opportunity identified in this analysis.

• Since service quality (86.4/100) is already strong and consistent, frame retention campaigns around convenience and rewards rather than service-quality messaging.

4. Payments & Operations

• Continue prioritizing EasyPaisa and digital-wallet reliability, since digital payments already make up 75.1% of revenue.

• Track Avg. VTAT (8.46 min) and CTAT (29.15 min) as ongoing operational-efficiency KPIs, segmented by vehicle type and zone, to catch dispatch or routing slowdowns early.

5. Reporting & Monitoring

• Extend the dashboard's date range in future refreshes to track whether the flat monthly trend holds as more data becomes available.

• Add a dedicated cancellations page filtered by reason and vehicle type so operations teams can act directly on the root causes identified above.

Business Impact

• Reducing driver-side cancellations from 18.01% toward industry-typical levels would recover thousands of otherwise-lost bookings per year.

• Lifting the 2.45% customer repeat rate even modestly would compound directly into revenue growth without new customer-acquisition spend.

• Sustaining the 86.4/100 service quality score while fixing cancellations would strengthen Bykea's competitive position in Karachi's ride-hailing market.

- A stable, well-understood monthly demand pattern supports more confident driver-supply and revenue planning.

Dashboard Highlights

The accompanying Power BI file (Bykea_Ride_Analysis.pbix) implements this analysis as a three-page interactive dashboard:

• **Home Page** — KPI cards for Total Booking, Completion Rate, Total Revenue and Avg. Booking Value; a booking-trend area chart; a Booking Status pie chart; a Vehicle Type pivot table (bookings, revenue, cancellations); and a Payment Method column chart, with slicers for Payment Category, Vehicle Type, Day Part and Date.

• **Revenue Page** — Cancellation-reason pie charts for both customer and driver cancellations, a Date-vs-Ride-Distance column chart, and a Vehicle Type pivot table comparing average Customer and Driver ratings.

• **Summary Page** — Custom KPI cards for Customer Retain Rate, Revenue per KM and Premium Peak Hour Index; a Monthly Revenue area chart; a Service Quality Score gauge; and clustered column/bar charts breaking down Booking Value by Day Part and Revenue Tier.

My Contribution to This Project

Beyond building the dashboard visuals, the bulk of this project's effort went into cleaning a genuinely messy 153,000-row raw export and building the analysis logic behind every KPI. This section documents that work.

A. Data Cleaning (Power Query / Python)

• Stripped duplicate wrapping quotes from Booking ID and Customer ID fields (e.g. converting """CNR5884300""" to CNR5884300).

• Removed 4,233 duplicate Booking ID records to avoid double-counting.

• Standardized inconsistent text casing and stray whitespace across Vehicle Type, Booking Status and Payment Method (e.g. 'CHINGCHI', ' chingchi ', 'Chingchi' → one canonical 'Chingchi' category) — 35 raw Vehicle Type variants collapsed into 7 clean categories, and 25 raw Booking Status variants collapsed into 5.

• Converted 'null' string placeholders in Payment Method into true missing values, distinguishing genuinely unpaid/non-completed bookings from data-entry gaps.

• Parsed inconsistent Date and Time formats into proper date/time types; 3,000 records had missing dates and were flagged rather than dropped, to preserve them for non-date-dependent analysis.

• Cleaned and title-cased Pickup Location and Drop Location fields (over 150 distinct locations) for consistent grouping and charting.

B. Analysis Logic Behind Each KPI

Every KPI in this report and on the dashboard is backed by an explicit calculation, mirroring how each would be expressed as a DAX measure in the Power BI model:

```
Completion Rate = Completed Bookings ÷ Total Bookings
```

```
Revenue per KM = SUM(Booking Value) ÷ SUM(Ride Distance), completed rides only
```

```
Customer Retain Rate = Customers with >1 completed ride ÷ Total unique customers
```

```
Premium Peak Hour Index = Premium-tier bookings in Morning/Evening ÷ All Premium-tier bookings
```

```
Service Quality Score = AVERAGE(Driver Ratings, Customer Rating) rescaled to a 0–100 index
```

Supporting derived fields — Day Part (Morning / Afternoon / Evening / Night, from the Time column), Payment Category (Cash vs Digital), and Revenue Tier (Low / Mid / High / Premium, from Booking Value) — were engineered specifically to power the slicers and breakdowns used throughout the Home, Revenue and Summary dashboard pages.

C. Visualization & Design

Built KPI cards, an area trend chart, pie charts for status and cancellation reasons, pivot tables, clustered column/bar charts, and a gauge for Service Quality Score. Applied a consistent navy-and-gold theme across all three pages with Date, Vehicle Type, Day Part and Payment Category slicers for interactive filtering.

Conclusion

The Bykea Ride Analytics dashboard shows a service processing a healthy Rs 47.26M in completed-ride revenue over 2024 on a base of 153,000 raw bookings, with Rickshaw as the leading vehicle type and Morning/Evening commute windows driving 60% of revenue. The data cleaning work — normalizing 35 vehicle-type and 25 booking-status text variants, removing 4,233 duplicate records, and correcting malformed IDs, dates and payment values — combined with a full set of purpose-built KPIs, made it possible to build a dashboard that is both accurate and genuinely actionable.

The clearest opportunities ahead are cancellation reduction — driver-side cancellations alone account for 18.01% of all bookings — and customer retention, where only 2.45% of customers currently rebook despite consistently strong service ratings (86.4/100). Addressing these two areas, alongside sustaining driver supply during peak Morning and Evening windows, would meaningfully strengthen both completion rates and overall revenue growth.